

Tesla Valve Analysis

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Abstract—This project aims to analyze the performance of a Tesla Valve under laminar flow utilizing computational fluid dynamics in ANSYS. Simulations were performed for both forward and reverse flows and across Reynolds numbers ranging from 50 to 350. Diodicity was the defining measure in our results. At low Reynolds numbers, diodicity approximated 1. However, at higher Reynolds numbers, it increases significantly, reaching a value of 1.41 with a Reynolds number of 350. The addition of more repeating elements shows results of increasing diodicity with additional elements up to a point. Flow visualizations show the presence of recirculation and momentum redirection that is consistent with the valve’s intended design. A grid convergence study validates our baseline mesh as acceptable as it differs from the fine mesh by $\approx 3\%$. Finally, design modifications were explored, with shortening the baffle length shown to produce the most significant improvement in diodicity.

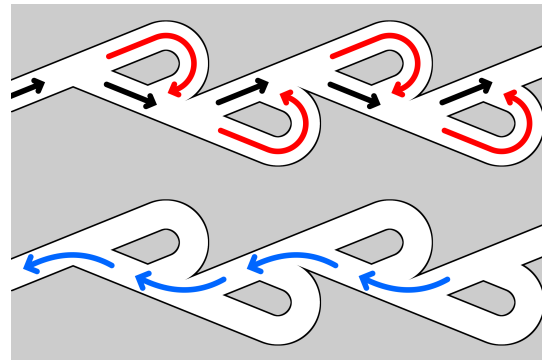


Fig. 1. Basic Tesla Valve performance.

NOMENCLATURE

Re	Reynolds number (dimensionless)
D	Diodicity
ΔP_{rev}	Reverse Flow Pressure Drop
ΔP_{fwd}	Forward Flow Pressure Drop
W	Characteristic Channel Width

I. INTRODUCTION

Fluid mechanics involves the study and control of fluid flow through applications such as containers, valve geometries, human systems, and many other natural and engineering systems. Such are applications of the Tesla Valve that can allow for a simple design for unidirectional flow. Tesla Valves have recently been explored due to its applications in micro fluids. At the Reynolds numbers we studied, ranging from 50 to 350, the flow remains laminar, allowing the role of viscous losses and geometric asymmetries to be isolated. [1].

In 1920, Nikola Tesla patented his version of a valve to allow fluid to pass with ease when flowing in one direction while the fluid experiments with significantly harder resistance when trying to pass in the opposite direction. Figure 1 shows the basics behind the Tesla Valve’s working principle. The effectiveness of the valve is measured by comparing the pressure losses in the forward and backward directions. Therefore, we used diodicity to quantify performance. It is especially interesting as it does not involve moving parts unlike many valves that are currently in use.

In this study, we analyzed laminar flow through a tesla valve using ANSYS fluent. Forward and reverse pressure losses are derived across changing Reynolds numbers. Diodicity quantifies performance while a grid convergence study validates our results.

A. Diodicity

The performance of the Tesla Valve is quantified by its diodicity which measures how effective the valve is at impeding reverse flow relative to forward flow. Per equation [1], diodicity is the ratio of the pressure drop in the reverse direction (ΔP_{rev}) to that in the forward direction (ΔP_{fwd}) (for the same flow rate). A higher diodicity corresponds to a better Tesla Valve performance, as it would mean the valve’s reverse flow would require a larger multiple of the pressure drop of the forward flow to achieve the same flow rate.

$$D = \frac{\Delta P_{rev}}{\Delta P_{fwd}} \quad (1)$$

Research has been done for the analysis of Tesla Valves. Typical values for diodicity of low Reynolds numbers are approximately $\approx 1 - 1.5$ which agree with our results. For Reynolds numbers of $Re = 500$, experiments have shown a diodicity of $D \approx 2.0$ with other experimental results varying. Differences in results “may be due to differences in geometry, both in that the diodes are variations on Tesla’s original design and the measurement systems may not isolate the channel, thereby introducing additional pressure drops that dilute diodicity” [2]. At higher Reynolds numbers, our results are likely to diverge from experimentally derived values. This is due to our simulation making use of over-idealized boundary conditions where small changes in geometry can have a significant impact.

B. Reynolds Number

The Reynolds Numbers that have been studied are relatively small Reynolds Numbers that correspond to laminar flow $Re < 2300$. This project studies Tesla Valve simulations at Reynolds Numbers between $50 \leq Re \leq 350$.

Simulations have shown that at $Re = 50$, diodicity is only very slightly above 1 [2] [3]. However, Nguyen et al. found that "diodicity turns on abruptly at Reynolds number $Re \approx 200$ " where some turbulence and instability can be seen [2]. The result is a sharp rise in diodicity, which is consistent with our data. As the Reynolds number increases, inertial effects become more prominent relative to viscous forces. This allows for flow separation, enhancing recirculation.

Liosis et al., *Microsystems* (2025) presents their diodicity results against different Reynold numbers in Table A2 [3] which we use to compare our results.

$$Re = U_{in} \frac{\rho W}{\mu} \quad (2)$$

C. Geometry

Most experimentation involving Tesla Valves vary in their results due to differing geometries all relying on the same principle. Multiple valve stages (repeating elements) in series can be set up to increase diodicity since each stage contributes additional resistance in the reverse direction while doing little to stop the forward flow. Liosis et al., *Microsystems* (2025) also compares changes in diodicity with additional stages [3].

II. MATERIALS AND METHODS

This study was conducted by using computational fluid dynamics (CFD) to analyze various laminar flows through a standard Tesla Valve. Simulations were performed on ANSYS Fluent software to derive the pressure drops in both the forward and reverse directions.

Air was set as our working fluid, with default density ($\rho = 1.225 \frac{kg}{m^3}$) and viscosity ($\mu = 1.7894 \times 10^{-5} \frac{kg}{m \cdot s}$). These values help set our inlet velocity to account for the required Reynolds Number per equation [2].

A baseline tesla valve was primarily used in our analysis. To iterate through different Reynolds numbers, we changed inlet velocity while keeping all other properties constant.

A grid convergence study was performed to show that our baseline mesh could act as a realistic simulation of a tesla valve. Flow fields of pathlines, pressure, and velocity were analyzed to ensure the valve was behaving as intended.

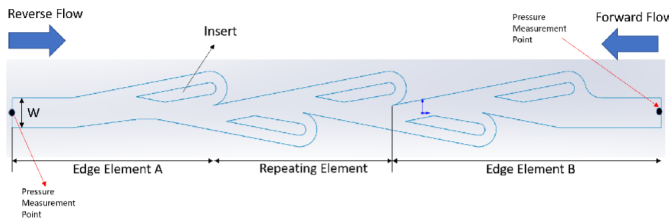


Fig. 2. Design schematic for baseline Tesla Valve.

1) *Simulation Geometry*: The geometry for our baseline testing and analysis was assembled using SolidWorks modeling software. A 2D plane was generated from a sketch of the converted entities from the assembly. This 2D plane was imported to ANSYS as a STEP file to serve as the simulation's geometry.

2) *Simulation Mesh*: A uniform mesh was used to account for the sharp gradients and parabolic profiles as well as between the inserts and side walls.

The baseline mesh had a 0.002 m element size with any automatic configurations (such as detect curvature, capture proximity, adaptive sizing, mesh defeaturing) turned off. The sizing feature was set with *hard* behavior.

III. RESULTS AND DISCUSSION

A. Baseline Simulation

We determined the inlet velocity, based on the given Reynolds number, the characteristic width of the pipe determined by the geometry, and the fluid characteristics of air, determined by the ANSYS software. Using equation [2] the inlet velocity for $Re = 350$ was determined to be $U_{in} = 0.2556 m/s$.

To simulate flow in the forward direction, the right wall was set as a *velocity_inlet* and the left was set as a *pressure_outlet*. The opposite boundary conditions were set to simulate flow in the reverse direction. The calculation was run using the predetermined values and a solution was converged.

The pressure drop was determined by calculating the Surface Integral Area Weighted Average of static pressure at the inlet and outlet and calculating the difference between them. Since the outlet was set as a pressure outlet with a pressure of 0 Pa, the pressure drop comes directly from the inlet pressure.

The forward pressure drop was determined to be $P_{fwd} = 0.2242111$ Pa and a reverse pressure drop was observed as $P_{rev} = 0.3166738$ Pa. From equation [1] give a calculated diodicity estimate of $D = 1.412391269$.

Our results show that the valve does in fact behave as per Tesla's intentions. This can be primarily seen from the pathlines and velocity profiles as shown in Figure [3]. From the forward case contour plots, we can see the velocity profile concentrated at the center of the tesla valve with less particles flowing through the insert channels, especially near the outlet. For the reverse case, the particles take the insert exits and flow back and counter and resist the main flow.

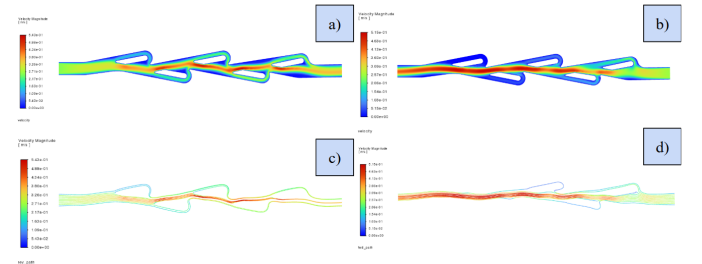


Fig. 3. a) Reverse velocity profile. b) Forward velocity profile. c) Reverse flow pathlines. d) Forward flow pathlines.

A grid convergence study was also made to compare the results of our baseline grid with a finer and coarser mesh. A mesh comparison is shown in Figure [4]. The mesh statistics were used to determine the number of elements in the baseline mesh and modify it to achieve the $\pm 50\%$ grid points.

Results are shown in Table [1]. The fine mesh per Figure [4c] should be our most accurate mesh, but the most computing

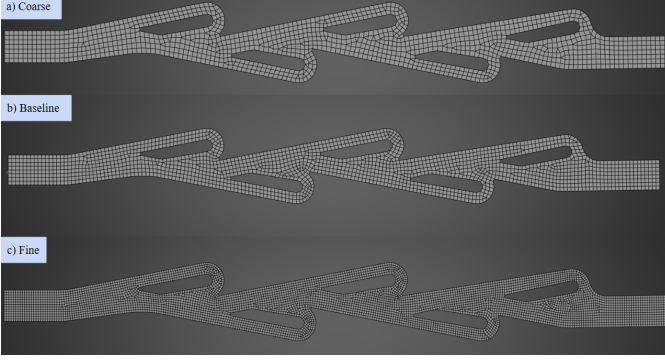


Fig. 4. Comparison between meshes for convergence study. a) Coarse mesh. b) Baseline mesh. c) Fine mesh.

intensive grid. Taking the fine mesh as our best available point of reference, we compared the coarse and baseline meshes to it. Focusing on diodicity, comparing the coarse mesh against the fine one, we get a percentage error of $(1.664 - 1.369)/1.369 \approx 21.6\%$ which is much larger than something we would want to use. However, the difference is much less apparent for the baseline mesh with a percentage error of $(1.412 - 1.369)/1.369 \approx 3.1\%$ relative to the fine mesh.

TABLE I
GRID CONVERGENCE STUDY RESULTS

Mesh	Elements	Diodicity	Element Size (m)
Coarse	1280	1.664	ES: 0.003 m
Baseline	2754	1.412	ES: 0.002 m
Fine	4824	1.369	ES: 0.001 m

Results for forward and reverse pressure drop as well as maximum peak velocity were also determined. Based on these results, we have taken our baseline grid to be acceptable and estimate the uncertainty of our baseline grid CFD results to be of $\approx 3\%$ for diodicity, $\approx 1.5\%$ for pressure drop (ΔP), and of $\approx 4\%$ for peak velocity; all acceptable values.

B. Reynolds Number Variations

A comparison of different Reynolds Numbers ($Re = 50, 100, 150, 250$) flowing through the Tesla Valve were also analyzed as part of this project. The air properties were kept constant ($\rho = 1.225 \frac{kg}{m^3}$, $\mu = 1.7894 \times 10^{-5} \frac{kg}{m \cdot s}$) while inlet velocity was varied to agree with the Reynolds Number being analyzed. Results are shown in Table II and a plot is shown in Figure 5.

TABLE II
VARYING REYNOLDS NUMBER RESULTS

Re	U_{in} (m/s)	ΔP_{rev} (Pa)	ΔP_{fwd} (Pa)	Diodicity
50	0.036525	0.0168	0.0172	0.975
100	0.07305	0.0392	0.0397	0.988
150	0.109575	0.0694	0.0671	1.034
250	0.182625	0.1609	0.136	1.178
(350)	0.2556	0.3166	0.224	1.412

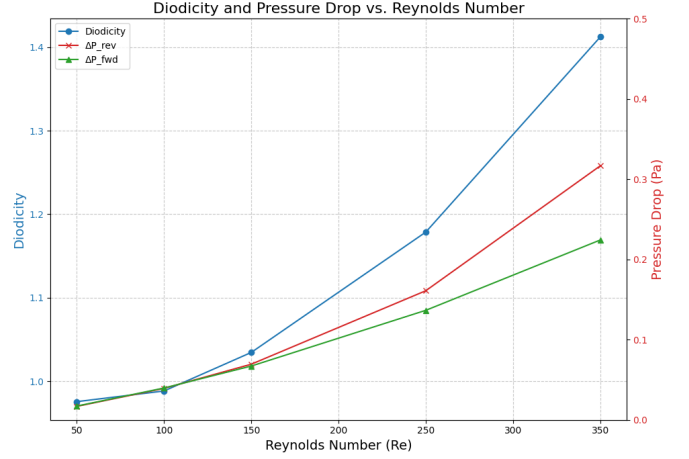


Fig. 5. Visual plot of diodicity and pressure drop against Reynolds Number.

Based on the idea that Reynolds Number represents $Re = (Inertial Forces)/(Viscous Forces)$, low Reynolds Numbers are those where viscous forces dominate and flow "sticks" together. Therefore, diodicity is very close to 1 since the flow is smooth in both directions.

It is important to note that the first couple of values are less than 1 which should not be possible per the Tesla Valve's purpose, but it is understandable given our previously discussed mesh uncertainty.

Our results agree with the literature review conducted. Liosis et al., Microsystems (2025) showed results of $D = 1.012, 1.053, 1.111, 1.227, 1.339$ for $Re = 50, 100, 150, 250, 350$ in a two-stage design which are very much in agreement with our results [3]. Our values might not be and could differ due to differences in geometry, meshes, analysis, as well as many other factors, but they are significantly close.

C. Repeating Elements

We also investigated the effect of adding additional repeating elements ($N = 1$ to 5) in series to the Tesla Valve to observe the net pressure drop in forward and reverse flow direction as well as calculated diodicity to quantify the valve's performance.

For each number of repeatable elements (N), the simulation was run using the same Reynolds Number case ($Re = 250$) to compare how adding repeating elements in series affects our study. Table III shares the results obtained by the simulations.

TABLE III
REPEATABLE ELEMENT STUDY RESULTS

N	ΔP_{rev} (Pa)	ΔP_{fwd} (Pa)	Diodicity
1	0.161	0.137	1.179
2	0.233	0.181	1.289
3	0.308	0.221	1.396
4	0.384	0.264	1.451
5	0.454	0.312	1.457

Figure 6 illustrates our pressure drop results for both the forward and reverse flow directions. This helps us understand

how increasing the number of elements results in a higher pressure drop difference between forward and reverse flow leading to a higher diodicity value.

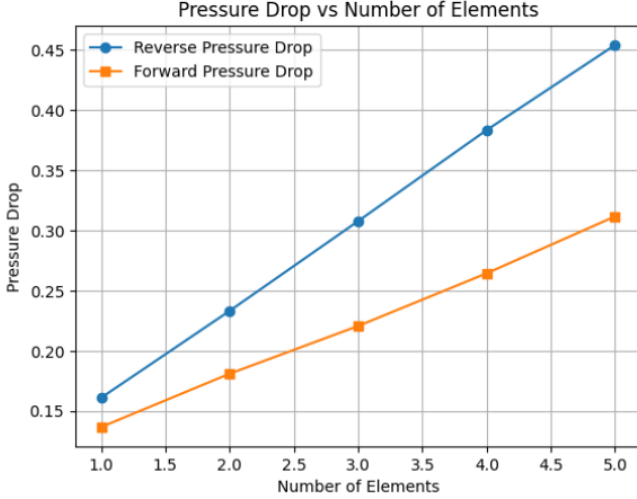


Fig. 6. Pressure drop for both forward and reverse flow plotted against the number of elements (from $N=1$ to $N=5$).

The net pressure drop increases linearly with the number of elements for both forward and reverse cases. The forward pressure drop has a slower rate of increase, showing the number of repeatable elements has a greater effect on the pressure drop going in the reverse direction.

The diodicity increases as the number of elements increases. This is because for each additional element, the additional resistance reverse flow encounters is greater than the relatively smaller resistance increase for forward flow. However, the diodicity per element decreases with the number of elements, showing that it reaches a saturation limit where diodicity will no longer increase and therefore converge. Figure 7 illustrates this in a plot.

Although additional repeating elements may help increase valve diodicity, the trade-off is a longer valve that may not fit a specific application.

D. Design Improvements

To improve the diodicity of the Tesla valve, we experimented with changing three variables in the geometry in the repeating unit.

First, we remodeled the repeating unit by increasing and decreasing the inner radius of each turn. Decreasing the radius creates less blockage for flow in the forward direction, which should decrease the pressure drop, but have minimal impact as fluid flows mainly in the primary channel. In reverse flow, a smaller radius means sharper turns and forces the fluid to collide more forcefully when entering the main flow path, increasing resistance and reverse pressure drop. Therefore, the diodicity from decreasing the inner radius is expected to increase.

Secondly, we lengthened and shortened the channel length, which is the length of the wall along a bend (baffle). We predict that decreasing the channel length will also create a

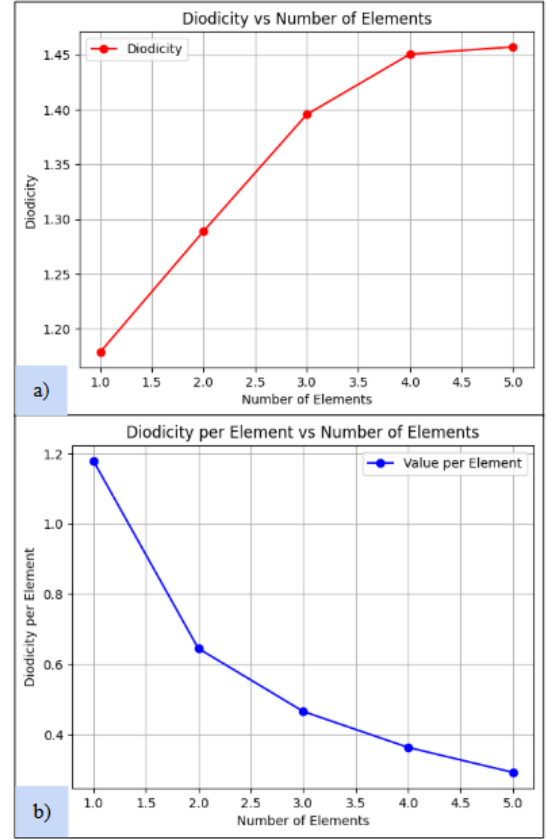


Fig. 7. Diodicity plotted against the number of elements. a) Total diodicity. b) Diodicity per element.

smoother path for the fluid to flow in the forward direction, because there is less momentum loss due to friction with the walls, which should decrease the forward pressure drop. In reverse, the fluid has less time to establish stable flow paths and encounters more sudden exits, which can lead to increased flow separation and mixing. Therefore, decreasing the channel length should result in a greater reverse pressure drop and the diodicity should increase.

Third, we shortened and lengthened the entry length, which is a straight section of each repeating unit before the fluid enters a bend. Shortening the entry length does not affect the geometry of each bend, but rather each element will shorten and each bend will become closer to a subsequent bend. We predict this to increase the resistance of reverse flow because it has a more abrupt turn between each bend, and improve the mixing of reverse flow entering the main stream. Wang et al. [4] found that shortening the valve to valve distance increased the diodicity of the Tesla valve in their experiment, and we wanted to test this finding.

For each design change until the final one, we only modified one dimension in the original model to understand how individual variables affected the pressure drop. Each change was simulated with three repeating units for consistency. The resulting diodicity and pressure drops were compared to our baseline results with three repeating elements at $Re = 250$, which is approximately 1.396. The pressure drops for that simulation are 0.308 Pa in the reverse direction and 0.221

Pa in the forward direction.

TABLE IV
DESIGN CHANGE STUDY RESULTS

Design Change	ΔP_{rev} (Pa)	ΔP_{fwd} (Pa)	Diodicity
Increased radius (0.6 cm)	0.248	0.227	1.091
Decreased radius (0.3 cm)	0.334	0.219	1.522
Increased channel length (4.96 cm)	0.336	0.250	1.342
Decreased channel length (3.96 cm)	0.306	0.211	1.454
Increased entry length (+1 cm)	0.288	0.222	1.300
Decreased entry length (-0.5 cm)	0.338	0.251	1.347
Decreased radius (0.15 cm)	0.362	0.225	1.611
Decreased channel length (3 cm)	0.459	0.210	2.182
Decreased radius (0.1 cm) and decreased channel length (3 cm)	0.320	0.177	1.809

Decreasing the channel length and inner radius were the two design changes that increased diodicity. To further explore these modifications, we analyzed models with even smaller radii and shorter channel lengths. As predicted, reducing the radius further decreased diodicity, while shortening the channel length to 3 cm produced the highest diodicity of approximately 2.182. This diodicity was higher than expected since it was the only of our trials to surpass a value of 2, so we remeshed the design with a 50% finer mesh (0.001 element size instead of 0.002) to better capture our geometry while keeping all other parameters constant to confirm our findings. After converging, the pressure drops were calculated to be 0.629 Pa in the reverse direction and 0.221 Pa in the forward direction, which is a diodicity of 2.847.

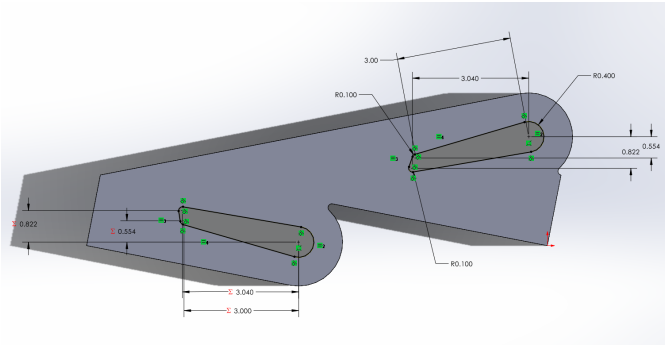


Fig. 8. Design modification sketch with channel length shortened to 3cm.

In the end, we combined our two promising design changes and created an even smaller radius and even shorter channel length design in the same model. However, the performance was difficult to predict due to changing multiple variables at the same time. It did increase diodicity compared to our baseline model, but at a lower level than just decreasing channel length as an individual change.

IV. CONCLUSION

This paper reviewed the performance of a Tesla valve in a laminar flow with Reynolds numbers ranging from 50 to 350. We made use of Computational Fluid Dynamics through ANSYS simulations where we analyzed the pressure drop in

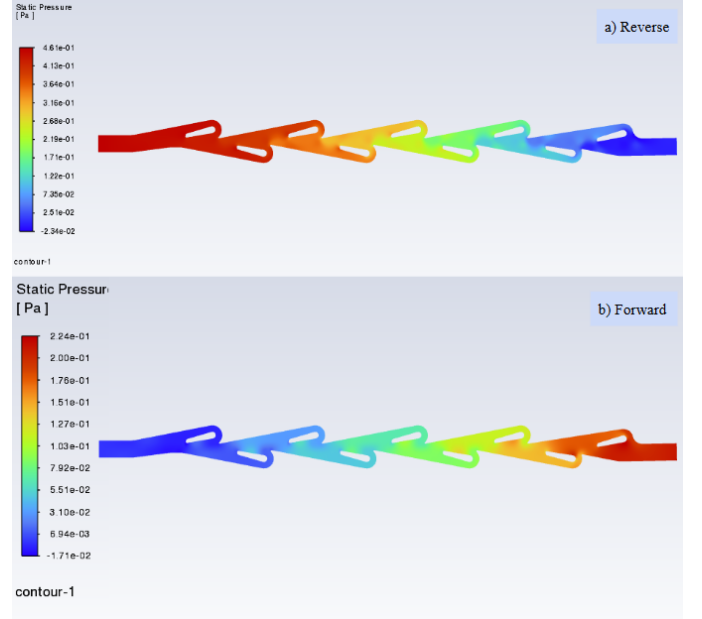


Fig. 9. ANSYS Pressure contour graph for 3 cm channel length design change. a) Reverse flow. b) Forward flow.

each direction. We derived the diodicity by dividing them, leaving us with the ratio that expressed efficiency. We ran the simulations with multiple different mesh qualities to perform a grid convergence study, and we were ultimately satisfied with the accuracy of the baseline mesh.

Our analysis revealed diodicity from 0.97 to 1.41, which lined up with values between 1 and 1.5 from our literature review. The relationship between diodicity and Reynolds number was not linear. When the Reynolds number was between 50 and 150, the diodicity remained practically unchanged, only increasing by .06. At $Re = 250$, the diodicity jumps up to 1.17, and at $Re = 350$, the baseline, it reached a staggering 1.41. This makes sense as this is a momentum-based valve. when the Reynolds number is lower, inlet velocity and initial pressure are lower, leading to viscous friction being the dominant force and rendering the loops on the valve useless. While we were happy with the accuracy of our results, future studies may expand their boundary conditions as they move into more turbulent regimes.

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